



Anamorphic imaging system for the moving viewer by projecting onto two orthogonal planes

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Abstract

The demands for stereoscopic images are rapidly growing in recent years. However, complicated system installation and expensive devices are usually required for those applications. Also, the restrictions of visual angles and the problem of slowly displaying the contents still exist. Furthermore, the traditional applications of anamorphosis are confined to fixed viewing positions with static content. Motivated by solving those problems, this paper presents an imaging system using one projector and perspective projection to generate static and dynamic anamorphic images. Then 3D images can be superimposed and displayed above two orthogonal planes. Furthermore, not only the images can be watched at a fixed position, but the function of demonstrating anamorphic images to viewers standing at different positions can also be achieved. Experimental results have shown that the proposed algorithms can successfully fulfil those two requirements. By using perspective projection to produce static and dynamic anamorphic images, this system can diversify the usage of stereoscopic images, provide high-quality images, and enhance the 3D visual effect. Also, those functions can increase the system variety in different applications of object demonstration.

Keywords Anamorphosis · Stereoscopic images · Perspective projection · Projector

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1 Introduction

Early in the fifteenth century, Greek artists and architects are the first to study *anamorphosis* [8]. This term comes from the Greek. The prefix *ana-*, means “back” or “again” and the word *morphe-* represents “shape” or “form.” Anamorphosis is a distorted projection or perspective requiring the viewer to occupy a specific point, use naked eyes (optical anamorphosis) or special devices like mirrors or lens (catoptric anamorphosis) to view a recognizable image [14]. The characteristic of anamorphosis is that it is hard to produce the content at first glance. Also, the embedded image can only be seen at a specific spot or using special equipment. The traditional anamorphosis is normally classified into two types [15]: perspective and mirror. A specific viewing point is defined by perspective anamorphosis [5] to view the distorted image. A classic example from Julian Beever’s work [1] is shown in Fig. 1. With mirror anamorphosis [11], a distorted image is painted on a plane, and the true image can be seen by placing a conical or cylindrical reflective mirror on the image. The cylindrical case of mirror anamorphosis is demonstrated in Fig. 2. Unlike perspective anamorphosis, catoptric images can be viewed from many angles [7].

The perspective anamorphosis can provide a 3-D visual effect without the help of any other auxiliary devices. Two common anamorphic videos with rich content had been produced by Kent [8]. Hansford and Collins [5] demonstrated how to make a 3D object by adopting anamorphic properties. However, although the perspective anamorphosis has been studied for a long time, there is still no further extension on other issues.

Due to the rapid development of imaging technology, the human demand to watch virtual things like real things grows increasingly. Therefore, the technology of 3-D display has been widely applied in many fields such as design, advertising, medical analysis, museum display, entertainment, and so on. With the application of Augmented Reality, anamorphosis can enhance the Visual Reality of virtual objects by providing a better three-dimensional display effect [4]. Other applications in this area include human-machine interactive editing of virtual objects [6], matching of building exterior mirror walls with the surrounding environment [3], positioning in the automated mechanical processes [9], the correction of imaging process to the masks in semiconductor industry [2], etc. However, for those 3-D display technologies, the following problems still need to be solved:



Fig. 1 Perspective anamorphosis [1]: (a) viewing at general positions and (b) viewing at the designated position



Fig. 2 Mirror anamorphosis [8]: **a** the distorted image and **b** the true image reflected on the cylindrical reflector

- (1) *Wearing auxiliary devices and employing expensive display equipment.* For most of the early 3-D vision technology, users were asked to wear special glasses. Based on the consideration of body reaction or sanitary factors, wearing external objects is unaccepted by most viewers. Hence, the naked stereoscopic vision technology arises in recent years that is expected to replace the method of wearing 3-D image viewing devices.
- (2) *Restricting the viewing range.* Some of the 3-D display technology confine the viewing range and the viewer's position. Without those limitations, the 3-D visual effect will disappear when the viewer departs from the designated area.
- (3) *Slow change to the display content.* Since the process to make the 3-D information content is extremely complicated, the display content is not allowed to be arbitrarily changed. Also, it is difficult to be dynamically adjusted corresponding to the moving position of the viewer.

Motivated by solving those three problems and extending the anamorphosis applications, this paper presents a novel 3-D display approach without attaching any auxiliary devices on the human body and an anamorphic imaging system is developed for a moving viewer by projecting onto two orthogonal planes. The proposed projection surface using two planes and system configuration is shown in Fig. 3. Since the traditional images made from perspective anamorphosis are painted on a single plane, a large area is required to show the 3-D effect (as depicted in Fig. 1a). As shown in Fig. 3a, we design a two-plane projection surface to overcome the space limitation. Figure 3b demonstrates the system configuration using a projector on the top to display the 3-D object content and a camera to simulate the viewer's eyes. Also, due to the limitation that the traditional perspective anamorphosis can only display the static content [1–9, 11, 14, 15], the proposed system will add a function of showing dynamic content by the projector to obtain a varied 3-D visual effect. Furthermore, to observe the best 3-D shape, the perspective anamorphosis must restrict the viewer to stand at a fixed position without moving. To remove this restriction, in this paper, a tracking mechanism to the viewer is designed to locate the viewer's position and the content is projected on the projection surface corresponding to this position.

Based on those functions, the projection methods of perspective anamorphosis are characterized by the types of the display content and the viewer's positions. The types of display

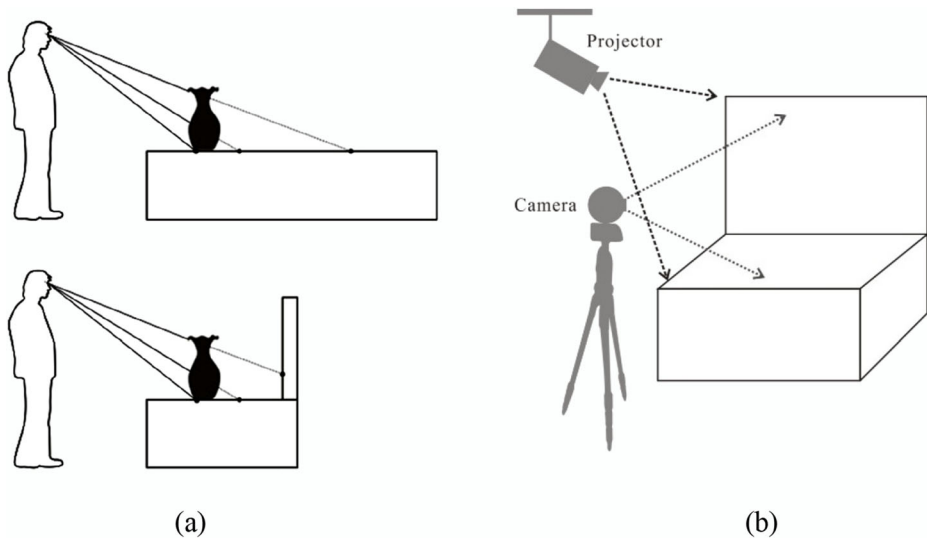
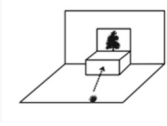
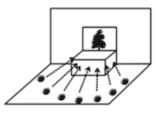
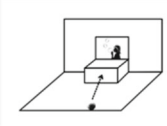
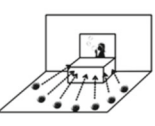


Fig. 3 **a** The two-plane projection surface and **b** system configuration

content projected by the projector can be classified into static images and dynamic videos. Furthermore, the viewer's positions are divided into fixed or varied. Therefore, the combination of categories can be tabulated in Table 1. By removing the display limitations of current anamorphic images, this paper presents a simple projection-based anamorphic imaging system. Using this system, the restrictions of displaying only static images and fixing the view angles are eliminated, and the proposed approach has provided a total solution. The proposed system is simple and lightweight. Also, complicated equipment and calculations are not required. In summary, the contribution of the proposed anamorphic imaging system includes the following items:

Table 1 The characteristics of the perspective anamorphosis projection methods

Viewer Content	Fixed	Varied
Static		
Dynamic		

- (1) Only one projector and two orthogonal planes are needed.
- (2) The types of display content can be static images or dynamic videos.
- (3) The viewer can change his viewing position along the display platform.
- (4) The display content can be changed quickly.

This paper presents an imaging system using one projector and perspective projection to generate static and dynamic anamorphic images. 3D images can be superimposed and displayed above two orthogonal planes. This paper is organized as follows. Section 2 introduces the preprocess of geometry and brightness calibration. Related algorithms to the proposed system are elaborated in detail in Section 3. Experimental results regarding how to generate static and dynamic anamorphic images are shown and explained in Section 4. Some conclusions are described in Section 5.

2 Geometry and brightness calibration

Since the proposed method uses only one projector to generate perspective anamorphosis images, geometry calibration, and brightness calibration between the projector and the camera simulating the human eyes need to be done first. Brightness calibration is used to solve the problem of brightness unbalance resulted from the projector to project the 3D content onto two orthogonal planes.

2.1 Planar Homography

In this paper, we adopt the World Coordinate System (*WCS*), which is a right-handed 3D Euclidean coordinate system in the 3D space. Three coordinate axes are given by X_W , Y_W , and Z_W . Figure 4 shows the 3D geometry configuration between the viewer(*C*) and the object(*W*). The camera is modeled as a pinhole camera and defined by another coordinate system, called the Camera Coordinate System (*CCS*). Three coordinate axes of *CCS* are given by X_C , Y_C , and Z_C . The origin of *CCS* is the center of projection (*COP*) of the camera, and Z_C is superimposed with the optical axis. The *CCS* will be employed to simulate the viewer position in which the optical axis is the line of sight, and *C* is the location of the viewer's head. Furthermore, the image plane inside the camera is expressed by a 2D coordinate system called the Image Coordinate System (*ICS*) defined by two axes of U and V . The normal vector of the image plane parallels with the Z_C axis of *CCS* and passes through the center of the image plane. The axes of U and V are parallel to X_C and Y_C , respectively. The distance between the image plane and the *COP* is the focal length f .

Perspective geometry can be adopted to create the mapping from the points on a plane to their corresponding points on another plane. This is also known as the plane projective transformation, the plane-to-plane projection, or the 2D-2D projective mapping [12]. This mapping can be represented by a 3×3 non-singular matrix which defines the mutual correspondence between two planes. The mapping relationships among those three coordinate systems can be found from the projection theorem [12]. The correspondence between the points on the input image (camera) and their corresponding points on the projection surface can be calculated. Therefore, based on the mapping, the input image can be corrected and projected onto the projection surface by referring to the standing position of the viewer.

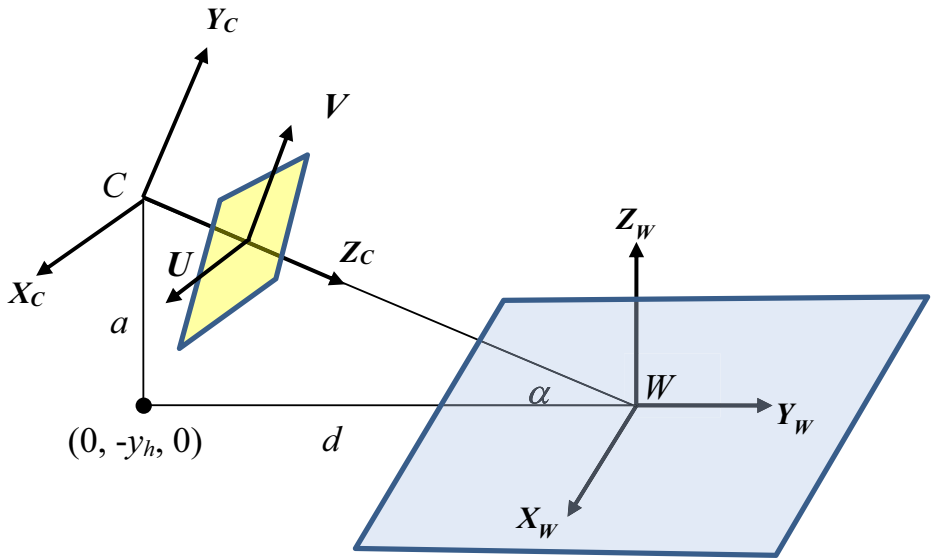


Fig. 4 3D geometry configuration between the viewer(C) and the object(W)

Assume that the point $P_W(X_{WP}, Y_{WP}, 0)_{WCS}$ is located on the projection plane, and the corresponding point of P_W on the image plane is $p(u, v)_{ICS}$ which can be described as

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = \begin{bmatrix} h'_{11} & h'_{12} & h'_{13} & h'_{14} \\ h'_{21} & h'_{22} & h'_{23} & h'_{24} \\ h'_{31} & h'_{32} & h'_{33} & h'_{34} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} = H \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} \quad (1)$$

where w is a scale factor, and H is a 3×3 homographic matrix. The coordinates of p are given by

$$u = \frac{wu}{w} = \frac{h_{11}X_w + h_{12}Y_w + h_{13}}{h_{31}X_w + h_{32}Y_w + h_{33}} \quad \text{and} \quad v = \frac{wv}{w} = \frac{h_{21}X_w + h_{22}Y_w + h_{23}}{h_{31}X_w + h_{32}Y_w + h_{33}} \quad (2)$$

Since the solution in homogeneous vector form is relative, we can set $h_{33} = 1$ to simplify the homographic matrix. Now consider four known corners, $P_i: (X_i, Y_i, 0)$, $i = 1, 2, 3, 4$, and their correspondent points $p_i: (u_i, v_i)$ define the display area on the projection plane and on the image plane, respectively. Equation (3) can be derived from Eq. (2) given by

$$\begin{bmatrix} X_1 & Y_1 & 1 & 0 & 0 & 0 & -u_1X_1 & -u_1Y_1 \\ 0 & 0 & 0 & X_1 & Y_1 & 1 & -v_1X_1 & -v_1Y_1 \\ X_2 & Y_2 & 1 & 0 & 0 & 0 & -u_2X_2 & -u_2Y_2 \\ 0 & 0 & 0 & X_2 & Y_2 & 1 & -v_2X_2 & -v_2Y_2 \\ X_3 & Y_3 & 1 & 0 & 0 & 0 & -u_3X_3 & -u_3Y_3 \\ 0 & 0 & 0 & X_3 & Y_3 & 1 & -v_3X_3 & -v_3Y_3 \\ X_4 & Y_4 & 1 & 0 & 0 & 0 & -u_4X_4 & -u_4Y_4 \\ 0 & 0 & 0 & X_4 & Y_4 & 1 & -v_4X_4 & -v_4Y_4 \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{bmatrix} \equiv Mh = D \quad (3)$$

While four points p_i are not collinear, the matrix K is an 8×8 non-singular matrix and the vector h can be obtained by

$$h = M^{-1}D \quad (4)$$

After rearranging the elements of h , the homographic matrix H can be rewritten as

$$H = M^{-1}(P_i, p_i)D(p_i) \quad (5)$$

2.2 Geometry calibration between the camera and the projector

As shown in Fig. 3b, this paper adopts the camera to simulate the viewer's eyes. The images captured by the viewer are firstly projected from the projector to the projected plane and then reflected to the viewer's eyes (image plane of the camera). Therefore, the relationship between the projector and the camera needs to be established first.

By referring to Fig. 4, the Z_C axis points toward the origin W of the WCS , and the position of the viewer C is $(x_C, y_C, z_C)_{WCS}$. From WCS to CCS , the position transformation of the object at $O : \{P | P(x, y, z) \in WCS\}$ is given by

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}_{CCS} = \begin{bmatrix} R \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{WCS} + \begin{bmatrix} x_C \\ y_C \\ z_C \end{bmatrix}_{WCS} \quad (6)$$

To compute (6) easily, using the homogeneous coordinate to represent (6) becomes

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}_{CCS} = \begin{bmatrix} R & x_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}_{WCS} \quad (7)$$

Based on the viewing configuration in Fig. 4, we can assume that $X_W \parallel X_C$ and two planes of Y_W-Z_W and Y_C-Z_C are orthogonal. Let $C:(x_C, y_C, z_C)_{WCS} = (0, -d, a)$, where a is the height of the viewer and d is the horizontal distance to W . Under this geometric arrangement, the rotation matrix R from WCS to CCS can be reduced to only one parameter, α , which is the rotation angle of C around X_W axis. This rotation matrix can be written as:

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix} \quad (8)$$

Then the principal point (u_0, v_0) is defined as the intersection (in pixels) of the optical axis of the CCS and the image plane. The skew factor k_r defines the skewness between axes of U and V on the image plane. Furthermore, the parameters k_u and k_v are the horizontal and vertical pixel spacing (mm/pixel) of axes of U and V , respectively. The focal length is denoted by f . Therefore, those intrinsic parameters can be summarized as the following matrix:

$$K = \begin{bmatrix} k_u & k_r & u_0 \\ 0 & k_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

In summary, for each point (X, Y, Z) in the WCS , the 3D to 2D projective mapping can be written as:

$$\begin{bmatrix} w \cdot u \\ w \cdot v \\ w \end{bmatrix} = \begin{bmatrix} k_u & 0 & u_0 \\ 0 & k_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & -d \\ 0 & \sin \alpha & \cos \alpha & a \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{WCS} \quad (10)$$

Since the relationship of the right angle between the optical axes of X and Y can be achieved by the current camera manufacturing technology, to simplify the projection model and the calculation, the skew factor kr is set to zero and disappeared in Eq. (10). The second matrix containing f in Eq. (10) is a perspective projection matrix used to create the transformation from 3D world coordinates to 2D image coordinates. Equation (10) is a complete geometric model of perspective projection. The benefit of this model is to fulfill the requirement of quickly getting the positions on the camera image plane corresponding to those points on the 3D object surface. Then the input image to the projector can be generated by using the homography relationship between two planes. Therefore, Eq. (10) is an explicit representation which can help the processing of dynamic 3D objects and the fast modification to the lateral displacement of the camera coordinate system during the movement of the viewer.

Let $(u_0, v_0) = (0, 0)$, Eq. (10) can be reduced to a compact form represented by

$$\begin{bmatrix} w \cdot u \\ w \cdot v \\ w \end{bmatrix} = \begin{bmatrix} k_u & 0 & 0 & 0 \\ 0 & k_v \cos \alpha & -k_v \sin \alpha & -dk_v \\ 0 & \sin \alpha / f & \cos \alpha / f & \alpha / f \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{WCS} \quad (11)$$

To calculate $[u \ v]^T$, Eq. (11) can be simplified and given by

$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{f}{Y \sin \alpha + Z \cos \alpha + a} \begin{bmatrix} X k_u \\ Y k_v \cos \alpha - Z k_v \sin \alpha - dk_v \end{bmatrix} \quad (12)$$

By Rearranging (12) yields

$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{f}{Y \sin \alpha + Z \cos \alpha + a} \begin{bmatrix} k_u & 0 \\ 0 & k_v \end{bmatrix} \begin{bmatrix} X \\ Y \cos \alpha - Z \sin \alpha - d \end{bmatrix} \quad (13)$$

Assume that two intrinsic parameters $k_u = k_v = k$ and other parameters f , a , d , and $\alpha = \tan^{-1}(a/d)$ are known, the imaging location of the point (X, Y, Z) is given by

$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{f \cdot k}{Y \sin \alpha + Z \cos \alpha + a} \begin{bmatrix} X \\ Y \cos \alpha + Z \sin \alpha - d \end{bmatrix} \quad (14)$$

2.3 Brightness calibration between the camera and the projector

As shown in Fig. 5a, after the image is projected onto a plane, since the brightness reflection is smoothly increased or reduced, the brightness of the reflected image captured by the viewer (camera) is near uniform. However, in the case of the projection surface with two orthogonal planes shown in Fig. 5b, since the brightness reflection from two different planes varies, the difference of brightness reflectance will be generated, particularly from the intersection of two planes. This will seriously cause the degradation of the realistic visual effect. Therefore, brightness calibration is needed and proposed to solve this problem. When brightness

degradation is caused by geometric factors, correction and calibration can be done by using the brightness reflectance model.

The Phong reflectance model [13] can be represented by

$$I = I_{\text{ambient}}(\text{ambient light}) + I_{\text{diffuse}}(\text{diffuse light}) + I_{\text{specular}}(\text{specular light}) \quad (15)$$

$$= I_a K_a + K_d (L \cdot N) I_d + K_s$$

in which N is the normal vector of point P above the surface, and the direction from P to the viewer is the vector CP . The vector from P to the light source is L . The vector r is the perfect reflection light of L . The strength of brightness reflection is related to the direction of three vectors: the viewing vector CP , the light vector L , and the surface normal vector N . Because two projective planes are orthogonal in this paper, the surface normal vectors, N_1 and N_2 from two surfaces, have a significant difference such that the reflection components along the viewing direction are different from two reflection planes. This will make CP a large change, and brightness calibration is needed before image projection by the projector.

Assume that P_1 and P_2 are reflection points on two orthogonal planes, respectively. N_1 and N_2 are their corresponding normal vectors. Two light vectors, L_1 and L_2 , are from the projector. The reflecting brightness captured by the viewer from two planes are depicted by

$$I_{d1} = I_1 k_d (L_1 \cdot N_1)$$

$$I_{d2} = I_2 k_d (L_2 \cdot N_2) \quad (16)$$

where I_i is the intensity of a point light source (i.e., the projector), and k_d is a wavelength-dependent empirical reflection coefficient. Brightness calibration aims to equal those two parameters of reflecting brightness. That is, $I_{d1} = I_{d2}$. Hence, $I_1(L_1 \cdot N_1) = I_2(L_2 \cdot N_2)$ and the ratio of reflecting brightness is given by.

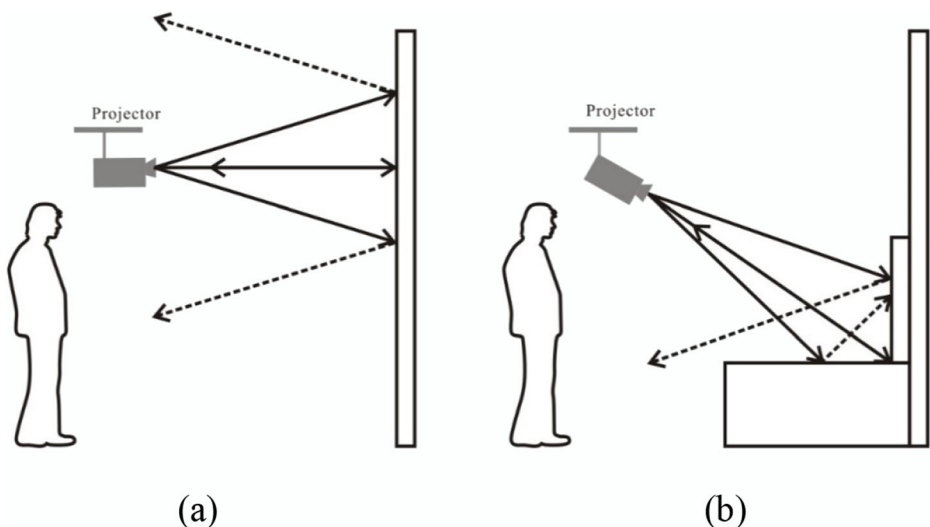


Fig. 5 Light Reflection of (a) Single Plane (b) Two Planes

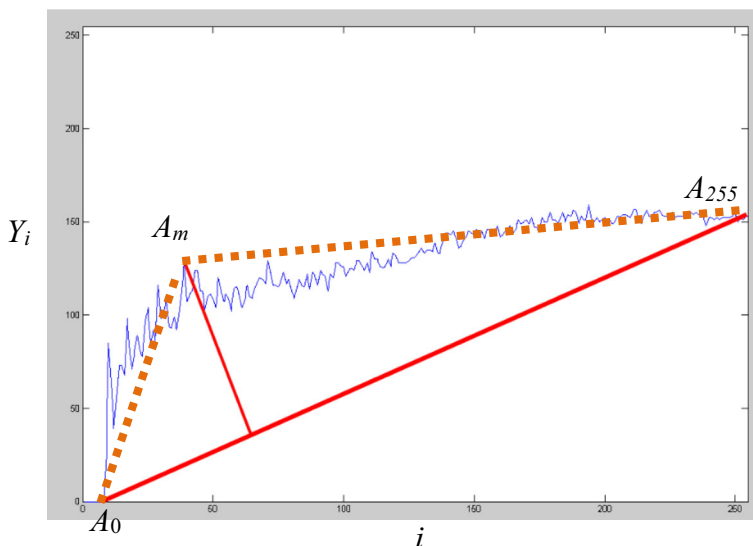


Fig. 6 The piecewise linear function for approximating the input-output response curve

$$\frac{I_2}{I_1} = \frac{L_1 \cdot N_1}{L_2 \cdot N_2} \quad (17)$$

This means that we can compensate the brightness of the projection surface using the scale factor based on a predetermined brightness reference.

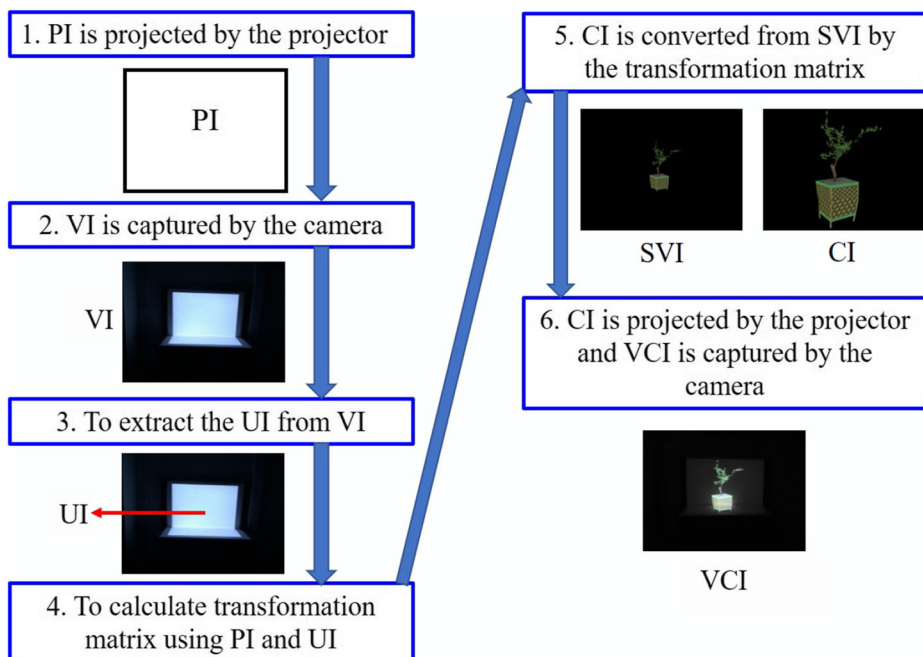
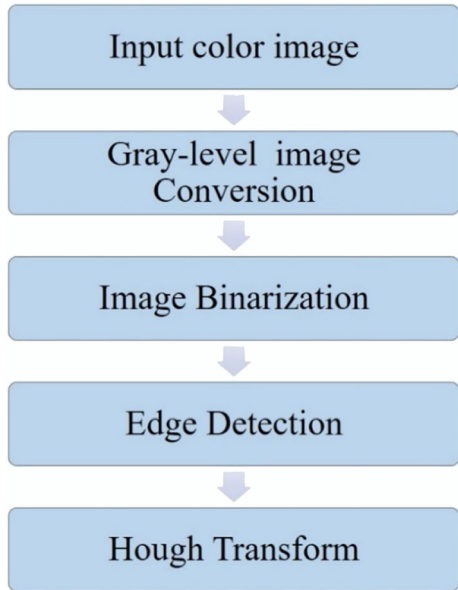
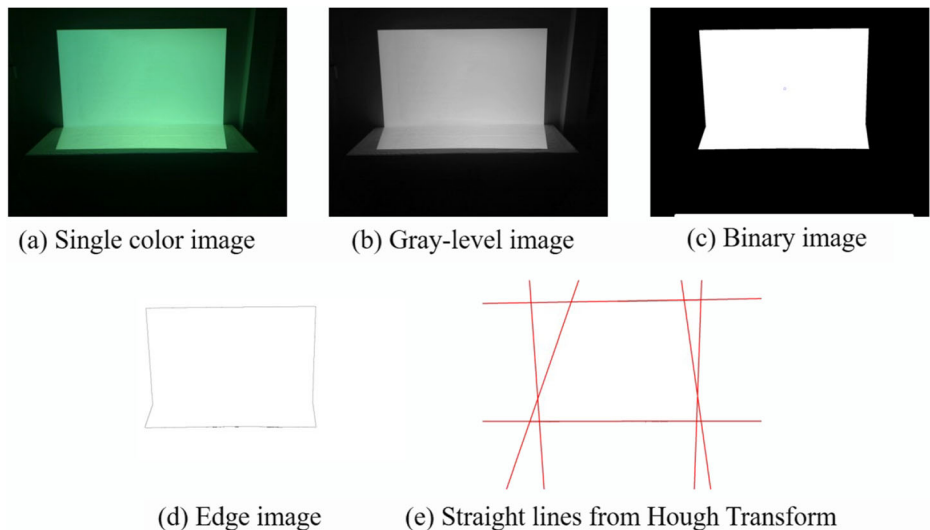


Fig. 7 The procedures of image calibration

Fig. 8 The flowchart of image preprocessing

In this paper, we have modified the brightness calibration method in [10] that only the brightest image is used to find the peak value of brightness from the sensing image. However, this method will make the brightness of all pixels ~~to~~ smaller than the peak value and generate a darker image. To tackle this problem, our technique employs 256 uniform calibration images with the intensities from 0 to 255 and records those 256 brightness responses as the intensity maps AM_0 – AM_{255} . For a pixel, its input-output response curve demonstrates the sensing brightness at different input brightness. Since the space required for storing the curve of each

**Fig. 9** The preprocessing results (a) Single color image (b) Gray-level image (c) Binary image (d) Edge image (e) Straight lines from Hough Transform

pixel is massive, a piecewise linear function approximation is applied to replace the complicated input-output response curve shown in Fig. 6. Therefore, brightness calibration can be performed by the following steps. First, assume that the point in the response curve is denoted as $B_i = (i, Y_i)$, $i = 0, \dots, 255$. The approximated curve is defined by two line segments, B_0B_m and B_mB_{255} , $m \in \{i\}$. The next step is to find the B_m such that the distance between the response curve and its approximation is the least. Let the line L is B_0B_{255} , and its equation is $ai + bY_i + c = 0$ where a , b , and c are the coefficients. For each B_i , the distance to L can be estimated by

$$d(B_i, L) = \frac{|ai + bY_i + c|}{\sqrt{a^2 + b^2}} \quad (18)$$

Hence, the optimal point B_m can be obtained by

$$m = \arg \max_{i, 1 \leq i < 255} d(B_i, L) \quad (19)$$

By using this algorithm, the system only needs to store three values, A_0 , A_m , and A_{255} , for each pixel, and other A_i values can be interpolated by the line segments of A_0A_m and A_mA_{255} .

3 Proposed scheme

The procedures of the proposed scheme are divided into image preprocessing, geometric calibration, and generation of the viewing image corresponding to the viewer's position. According to Table 1, the proposed scheme presents four projection methods. Before describing the proposed methods, the calibration between the camera and the projector needs to be done first. By referring to Fig. 3b, the procedures of image calibration are shown in Fig. 7.

During the calibration stage, the projector projects the image with a single color called Projector Image (PI). This image is used to extract feature points from the projection area. After this image is captured by the camera, which simulates the viewer's eyes, this image is called Viewing Image (VI). The image displayed inside the projection area is called the

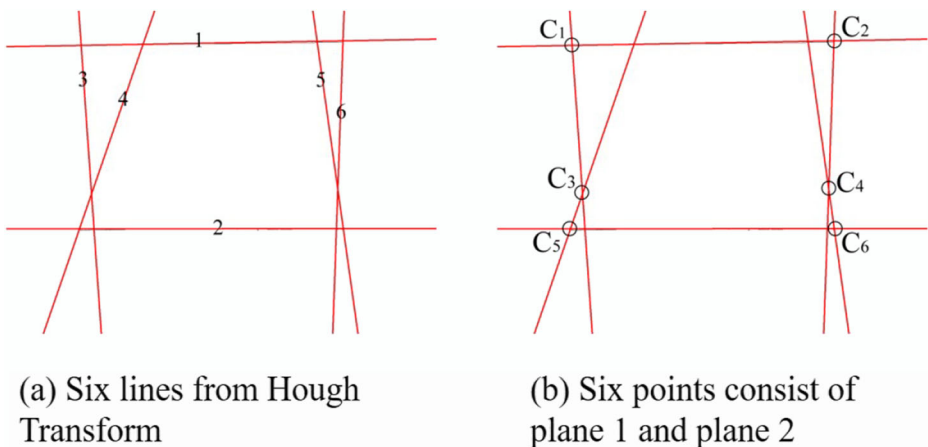


Fig. 10 Plane segmentation and representation (a) Six lines from Hough Transform (b) Six points consist of plane 1 and plane 2

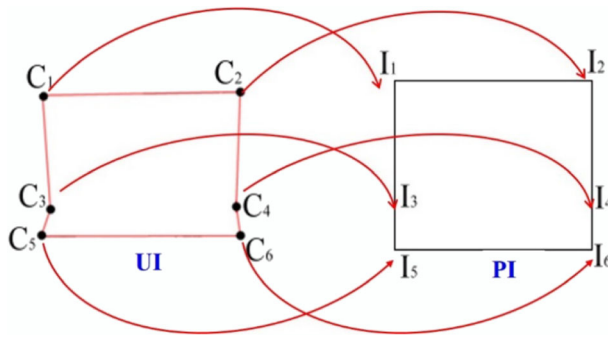


Fig. 11 Mapping relationship of feature points between UI and PI

Uncorrected Image (UI). By using the geometric relationship between PI and UI, the transformation matrix can be calculated, and the display content called the Simulated Viewing Image (SVI) can be converted into the Corrected Image (CI) by the transformation matrix. After this CI is projected by the projector, the image captured by the viewer (camera) is corrected, and this image is called the Viewing Corrected Image (VCI).

3.1 Displaying static content at a fixed viewing position

To display static content at a fixed viewing position, the geometric correspondence between the projector and the camera is adopted. Since geometric transformation is needed for the image captured by the camera, basic image preprocessing is performed. Then the plane segmentation is applied to extract the feature points for geometric transformation.

The aim of basic image preprocessing is to extract feature points from the image, and the flowchart is shown in Fig. 8. To reduce the computation complexity during image processing, the color image is converted into a gray-level image first. In this paper, YIQ color space is adopted, and the conversion from RGB color space to YIQ color space is represented by

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \quad (20)$$

in which the Y component denotes the gray-level value of each pixel. After that, to improve the computation efficiency, the gray-level image is usually converted into a binary image. Furthermore, edge detection is applied to this binary image, and the method of Sobel Edge Detection is used.

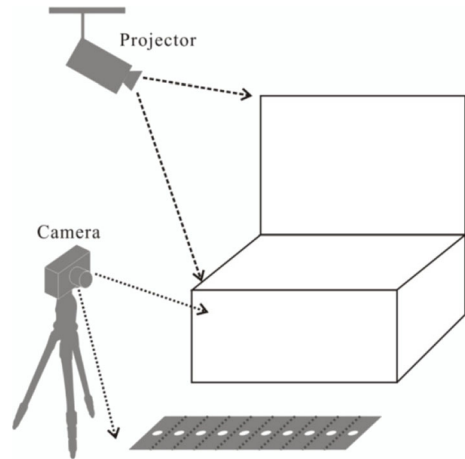
After edge detection, the Hough Transform is employed to extract straight lines from the image and those lines are used to determine the boundary of the projection area. As a feature extraction technique, the classical Hough transform is used to find straight lines by a voting procedure that is carried out in a parameter space. The first step is to map each point (x, y) along the straight line into the (ρ, θ) space and the line equation is defined by

$$x \cos \theta + y \sin \theta = \rho \quad (21)$$

in which ρ is calculated corresponding to different θ values in the range of ± 90 degrees. In the (ρ, θ) parameter space, straight lines exist in those positions with the accumulated number larger than a predefined threshold value. Figure 9 shows the preprocessing results in this paper.

For traditional techniques of corner detection, a default window is used to extract and analyze the specific internal signal. To determine the existence and locations of the corners,

Fig. 12 The system configuration for varied viewing positions



local information is usually adopted and only the positions of pixel coordinate with integer values can be obtained. However, based on the calculation of corner positions using the homographic matrix, the accuracy and the quality of projection pre-processing can be improved by using corner coordinates with subpixel precision. Therefore, instead of using traditional approaches, this paper adopts the projected image edges supported by multiple stable edge points. The corner positions with subpixel precision are determined by the intersection of lines from the Hough transform. Since the calibration of searching corners is an offline operation and is only performed once, even though the computation of Hough transform is more complicated than that of corner detection, this will not increase too much burden to the overall operations.

Fig. 13 The preprocess of viewpoint decision

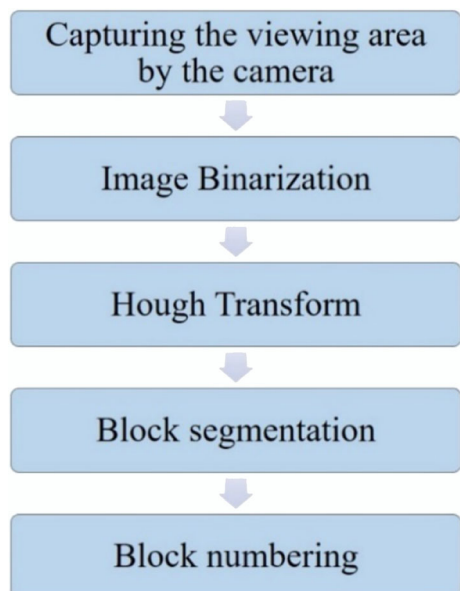
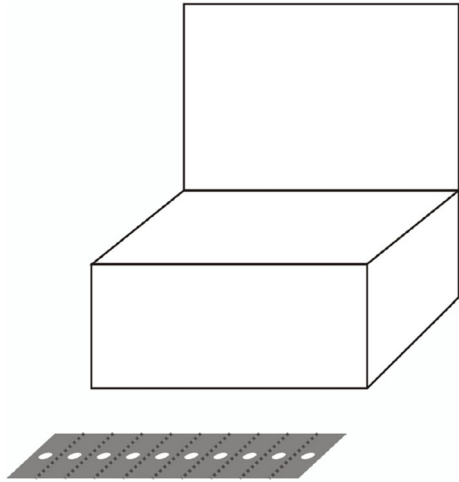


Fig. 14 The arrangement of the varied viewpoints



After basic image preprocessing, the plane segmentation can be done. Since two orthogonal planes are used in the proposed scheme as the projection planes, the projection area is covered by six lines. As shown in Fig. 10a, six straight lines from Hough Transform are obtained and ten intersection points are generated after the image preprocessing. However, the projection area shown in Fig. 10b consists of only six of those ten points. Therefore, we can set that plane 1 and plane 2 are constituted by those intersection points in the set of $C: \{C_1, C_2, C_3, C_4, C_5, C_6\}$. For those six lines in Fig. 10a, line angles can be adopted to decide the sequence during Hough Transform. The angles of line 1 and line 2 are both zero, and the Y value of line 1 is smaller than that of line 2. After line 1 and line 2 are confirmed, the lines intersected with line 1 are checked. For those lines, the line interests at the minimum X value of line 1 is line 3, and line 6 intersects at the maximum X value of line 1. Moreover, after checking the lines intersected with line 2, the line interests at the minimum X value of line 2 is line 4, and line 5 intersects at the maximum X value of line 2. After those derivations, the set of $C: \{C_1, C_2, C_3, C_4, C_5, C_6\}$ can be obtained in which plane 1 consists of (C_1, C_2, C_3, C_4) and plane 2 consists of (C_3, C_4, C_5, C_6) .

After plane segmentation, the geometric transformation between the projector and the camera is performed. On the UI image, six feature points can be extracted by the plane

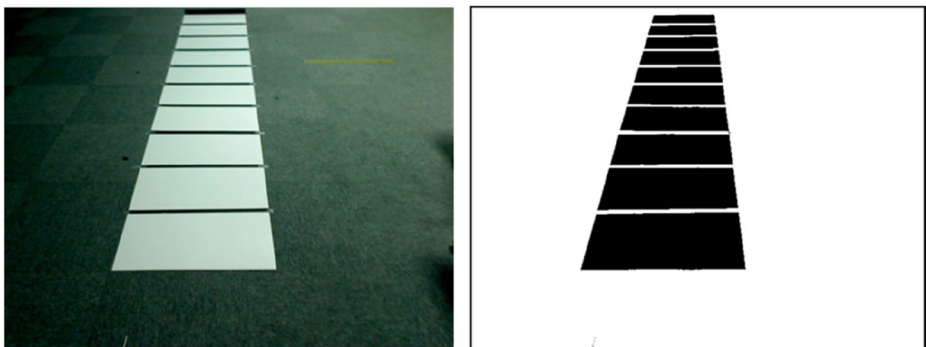
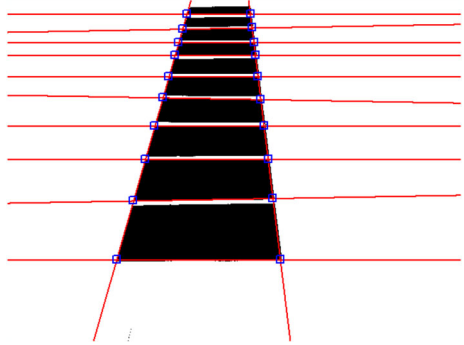


Fig. 15 The image of the moving area and the binary image

Fig. 16 The segmentation results of the moving area



segmentation. As shown in Fig. 11, the geometric transformation can be computed by using those six points from the set of C and their corresponding positions on the PI image. As described earlier, the projection area captured by the camera is divided into plane 1 and plane 2 by feature point extraction. Plane 1 is constituted by four points from $C_1: \{C_1, C_2, C_3, C_4\}$, and their corresponding points on PI are $P_1: \{I_1, I_2, I_3, I_4\}$. Plane 2 is constituted by four points from $C_2: \{C_3, C_4, C_5, C_6\}$, and their corresponding points on PI are $P_2: \{I_3, I_4, I_5, I_6\}$. Assume that the four points on the projection plane 1 are $P_1: (I_{ix}, I_{iy})$, and the feature points obtained from the image captured by the camera are $C_1: (c_{ix}, c_{iy})$ in which $i = 1, 2, 3, 4$. By referring to Eq. (3), the geometric transformation between P_1 and C_1 is given by

$$\begin{bmatrix} c_{1x} & c_{1y} & 1 & 0 & 0 & 0 & -I_{1x}c_{1x} & -I_{1x}c_{1y} \\ 0 & 0 & 0 & c_{1x} & c_{1y} & 1 & -I_{1y}c_{1x} & -I_{1y}c_{1y} \\ c_{2x} & c_{2y} & 1 & 0 & 0 & 0 & -I_{2x}c_{2x} & -I_{2x}c_{2y} \\ 0 & 0 & 0 & c_{2x} & c_{2y} & 1 & -I_{2y}c_{2x} & -I_{2y}c_{2y} \\ c_{3x} & c_{3y} & 1 & 0 & 0 & 0 & -I_{3x}c_{3x} & -I_{3x}c_{3y} \\ 0 & 0 & 0 & c_{3x} & c_{3y} & 1 & -I_{3y}c_{3x} & -I_{3y}c_{3y} \\ c_{4x} & c_{4y} & 1 & 0 & 0 & 0 & -I_{4x}c_{4x} & -I_{4x}c_{4y} \\ 0 & 0 & 0 & c_{4x} & c_{4y} & 1 & -I_{4y}c_{4x} & -I_{4y}c_{4y} \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} I_{1x} \\ I_{1y} \\ I_{2x} \\ I_{2y} \\ I_{3x} \\ I_{3y} \\ I_{4x} \\ I_{4y} \end{bmatrix} = C_1 h = P_1 \quad (22)$$

From (22), we can derive the result $h_1 = C_1^{-1} P_1$. Using this derivation, the image mapping for all pixels between PI and UI can be obtained by the following steps:



Fig. 17 The example of the viewer varied along the moving area

- Step 1: Using the mapping between $\mathbf{C}_1: \{C_1, C_2, C_3, C_4\}$ and $\mathbf{P}_1: \{I_1, I_2, I_3, I_4\}$ in (22), we can have $h_1 = C_1^{-1}P_1$ and rearrange h_1 to a 3×3 homographic matrix H_1 as shown in Eq. (1).
- Step 2: Using the mapping between $\mathbf{C}_2: \{C_3, C_4, C_5, C_6\}$ and $\mathbf{P}_2: \{I_3, I_4, I_5, I_6\}$, we can have $h_2 = C_2^{-1}P_2$ and rearrange h_2 to a 3×3 homographic matrix H_2 as shown in Eq. (1).
- Step 3: The inverse matrix of H_1 can be calculated and denoted by H_1^{-1} .
- Step 4: The inverse matrix of H_2 can be calculated and denoted by H_2^{-1} .
- Step 5: Using H_1^{-1} and H_2^{-1} , to display static content at a fixed viewing position can be achieved.

3.2 Displaying dynamic content at a fixed viewing position

For traditional perspective anamorphosis, the image content captured by the viewer is static and the geometric relationship between the image and the viewer is fixed when the viewer stands still. Since the projector can change the display content easily compared to the painted content, this merit is used by the proposed scheme to display dynamic content. Before displaying dynamic content at a fixed viewing position, the geometric transformation between the projector and the camera needs to be calculated and the lookup table (LUT) of transformation is created simultaneously. This LUT can be used to speed up the geometric transformation by only retrieving the position mapping without calculating the transformation again.

3.3 Displaying static content at varied viewing positions

The viewing area of the viewer is restricted by most of the 3D display techniques. Although some applications allow the viewer's position to move, the cost to pay is the degradation of image resolution. To solve the problems of viewpoint limitation and resolution degradation, this paper allows the viewer to watch the static content at varied viewing positions. To accomplish this requirement, the viewer's position needs to be known first. Then the 3D content can be displayed and captured in real-time corresponding to the viewer's specific position. As shown in Fig. 12, the proposed scheme uses a camera to track the viewer's position and this is different from the camera used during the calibration procedure to simulate the viewer's position.

Since most of the viewer's movement belongs to lateral displacement inside an exhibition space, the viewing area can be divided into several blocks. The corresponding display content appears when the viewer enters the designated block. Using this design, the display content for different blocks can be made in advance and the detection of viewpoint can be easily made. This paper uses the camera to capture the moving area of the viewer and track the viewer's position. The preprocess of viewpoint determination is shown in Fig. 13. To decide the viewpoint of the viewer, the range of the varied viewpoints needs to be defined and the decision is related to the viewing angle. The range of the viewing angle at each viewpoint can be decided by two factors:

- (1) *The size of the moving area for the viewer.* The viewing angle of the display content is increased with the size of the moving area for the viewer.

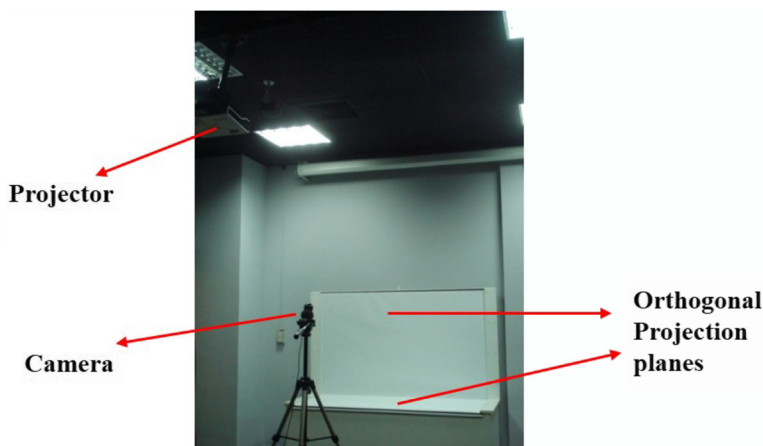
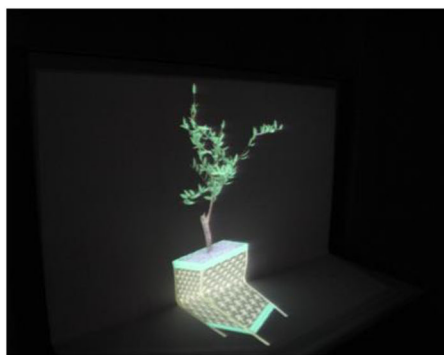


Fig. 18 Environment setup

- (2) *The number of the block segmentation.* The sense of viewing at fine segmentation is more consistent than that of rough segmentation. An insufficient number of blocks will result in an intermittent sense of viewing.

By considering those two factors, the aim of designing viewpoints is that the viewer is not aware of switching the display content while changing the viewpoint. As shown in Fig. 14, this range is set to a rectangle with a fixed width, and the length is 300 cm with every 30 cm a block. Each block represents a location to a specific viewpoint of the viewer, and there are ten different viewpoints.

Based on the factors determining the length and the width of the moving area described above, we can know that after the viewer has moved a certain distance and a visual difference to the projected object is generated, the viewing angle needs to be changed. Also, inside the range of the same viewing angle, the viewer is hard to tell the visual difference by the lateral movement. After repetitive experiments by the viewer, the projection range is divided into 10



(a) wrong viewpoint



(b) correct viewpoint

Fig. 19 Displaying the static content of a plant pot at two different viewpoints. **a** wrong viewpoint **b** correct viewpoint



Fig. 20 Displaying the static content at a fixed viewing position

blocks and each movement distance of the viewer is 30 cm. Therefore, inside the range of 3 m where the viewer can have lateral movement, the range is divided into 10 blocks and the width of each block is 30 cm.

According to the preprocessing steps in Fig. 13, the position of each viewpoint can be defined first by using the moving area of the viewer captured by the camera. The image of the moving area and the result of image binarization are shown in Fig. 15. After image

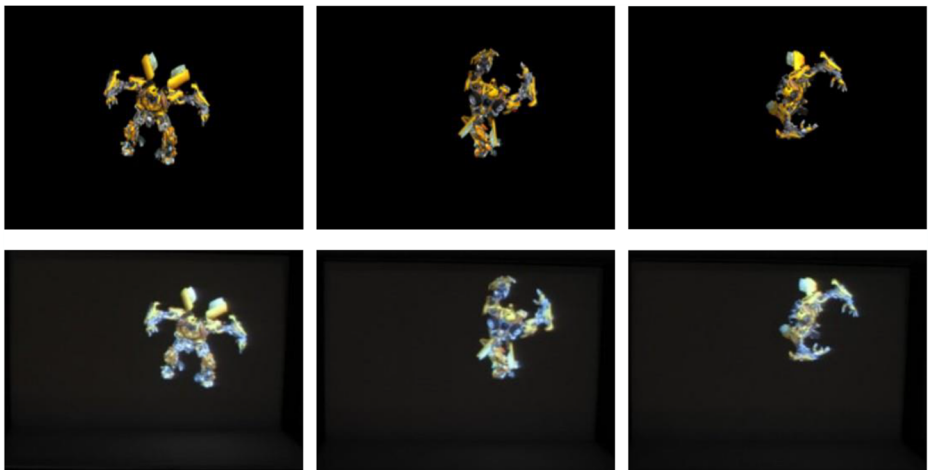


Fig. 21 The example of dynamic content with the motion of Bumblebee

binarization, as shown in Fig. 16, the lines segmenting all blocks can be found by using Hough Transform and the blocks are numbered from 1 to 10. After the viewer walks into the moving area, the proposed scheme starts to track the foot part of the viewer. Then the block number can be decided by scanning the white area from the bottom of the binary image, as shown in Fig. 17.

After the above-mentioned processing and arrangement, the requirement of displaying static content at varied viewing positions can be accomplished. As shown in Fig. 15, the sequence of white spots represents the track of moving viewpoints of the viewer. For each white area along the path, the lookup table (LUT) of the geometric transformation between the projector and the camera can be calculated off-line. Since the walking speed of a normal person is 1.25 m/s, the proposed scheme uses the camera to detect the viewer's position every 0.05 s. After the camera has detected the viewer's position and the block number is decided, the corresponding LUT can be retrieved and used to convert the uncorrected image. Then the corrected image is projected by the projector.

3.4 Displaying dynamic content at varied viewing positions

As described in Section 3.2, the approach of displaying dynamic content at a fixed viewing position requires the transmission of a massive amount of data. To display dynamic content at multiple viewpoints, much more data is needed. Therefore, the solution is to create all dynamic content for various viewpoints off-line. After that, the tracking of the viewer's position and the displaying sequence of dynamic content are operated online. Using the camera to detect the viewpoint of the viewer, the detection rate is every 0.05 s. When the viewer is detected to move again along the moving area, the sequence timer is simultaneously activated, and the corresponding 3D content at the new viewpoint is displayed. However, to keep the consistency, the content displayed at this new viewpoint must continue from the content displayed at the previous viewpoint. After the arrangement of those operations, the approach of displaying dynamic content at varied viewing positions can be accomplished.

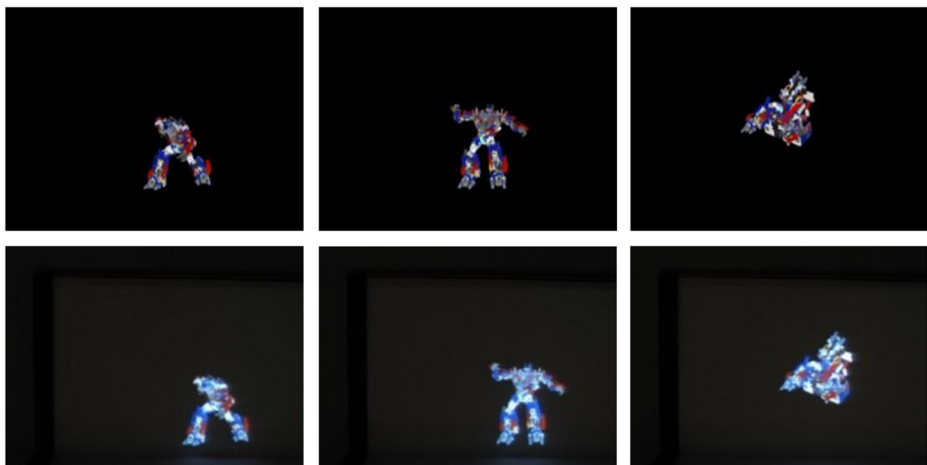


Fig. 22 The example of dynamic content with the motion of Optimus Prime

3.5 Discussions of computational efficiency

The discussions of computational efficiency for the proposed imaging system applied in four application cases are summarized as follows:

- (1) Watching the static object at a specific viewing position. Since the projected images are calculated beforehand, no extra computation and adjustment is needed during the projection. Therefore, the problems of processing time and resolution are inexistent.
- (2) Watching the dynamic object at a specific viewing position. Since the projected video is also calculated beforehand, no extra computation and adjustment is needed during the projection. Therefore, the processing time equals the time of video playing. There is no problem for the resolution.
- (3) Watching the static object by moving the viewing position. Since the system only needs to decide the viewer's position and the projected images are calculated beforehand, the only thing to do is to project the image corresponding to the viewer's position decided by the system. No extra computation and adjustment are needed during the projection and the real-time process can be accomplished.
- (4) Watching the dynamic object by moving the viewing position. The projected videos are calculated beforehand based on multiple viewing positions (10 blocks). After the viewer's position (block) is decided by the system, the corresponding video is projected based on the current viewer's position. Therefore, no extra computation and adjustment is needed during the projection. After the viewer moved from the previous position to another position, the projected video will switch to a corresponding video and the playing time continues from the previous playing. Therefore, the requirement of real-time processing and the smoothness of video playing are achieved.

4 Experimental results

Figure 18 shows the environment setup in the experiments. Two white boards are used as the orthogonal projection planes, and the projector is mounted on the ceiling. The camera is installed on top of a tripod. Two professional 3D computer graphics programs, MAYA and 3ds Max, are used for making the 3D models, simulation images, and videos. To demonstrate the performance of the proposed approaches, we have completed four experiments:

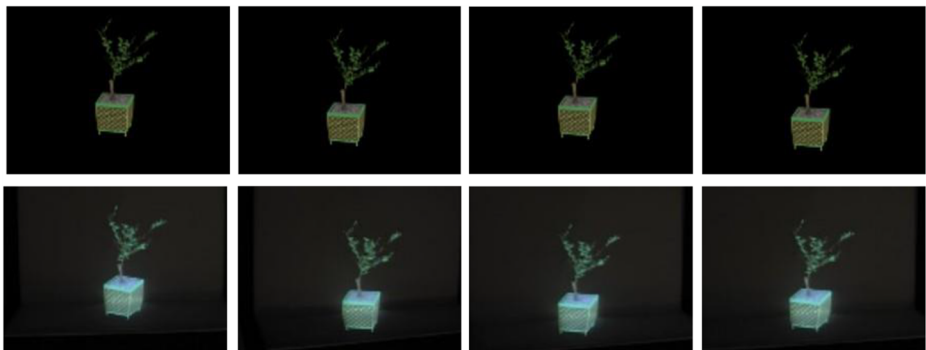


Fig. 23 The example of static content with a plant pot captured at different viewpoints

- (1) Experiment 1: watching the static content at a fixed viewing position
- (2) Experiment 2: watching the dynamic content at a fixed viewing position
- (3) Experiment 3: watching the static content at varied viewing positions
- (4) Experiment 4: watching the dynamic content at varied viewing positions

The dataset used in the experiments is generated by the second author using the MAYA and the 3ds Max software tools. The 3D images in this dataset include the plant and the car (Experiment 1), the Bumblebee and the Optimus Prime (Experiment 2), the plant pot (Experiment 3), and the running train (Experiment 4).

4.1 Experiment 1: Watching the static content at a fixed viewing position

For the viewer to watch the static content at a fixed viewing position, the example of static content with a plant pot captured at two different viewpoints is shown in Fig. 19. The accurate 3D content can only be captured from a specific viewpoint. Figure 20 shows another two examples of static content with a potted plant and a model car. Two SVIs are placed in the first row, and two corresponding VCIs are listed in the second row.

4.2 Experiment 2: Watching the dynamic content at a fixed viewing position

Figure 21 shows the first example of dynamic content with the motion of Bumblebee. Three sequential SVIs with an interval of 20 frames are placed in the first row, and three

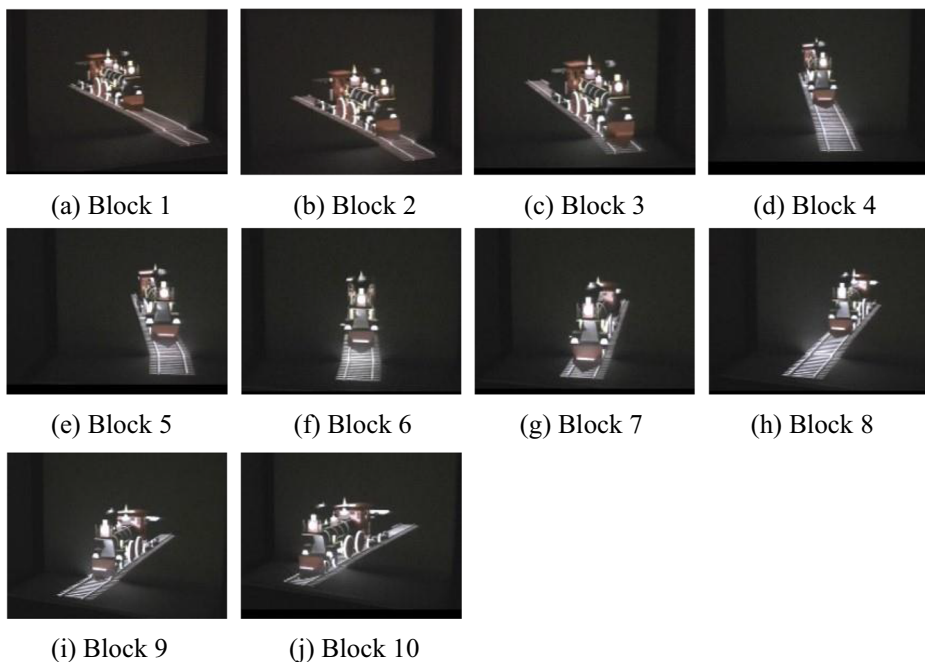


Fig. 24 The example of dynamic content with a moving train captured at different viewpoints. (a) Block 1 (b) Block 2 (c) Block 3 (d) Block 4 (e) Block 5 (f) Block 6 (g) Block 7 (h) Block 8 (i) Block 9 (j) Block 10

corresponding VCIs captured by the camera are listed in the second row. Figure 22 shows the second example of dynamic content with the motion of Optimus Prime.

4.3 Experiment 3: Watching the static content at varied viewing positions

For the viewer to watch the 3D content at varied viewing positions, Fig. 23 shows the example of static content with a plant pot captured from different viewpoints. A sequence of SVIs captured from adjacent viewpoints is placed in the first row, and their corresponding VCIs are listed in the second row.

4.4 Experiment 4: Watching the dynamic content at varied viewing positions

For the case of displaying dynamic content at varied viewing positions, the example shows a train is moving from far to near that like pass through the screen. The sequential images sampled at each viewpoint can be generated, as shown in Fig. 24.

Since the main purpose of Experiment 1 is to demonstrate the display performance of the static object, the input image with high resolution is adopted as the object to be projected. Therefore, the resolution of the projected image is better than that of other methods. For the other experiments, since the processed object is dynamic, during the stage of model building and projection processing, the input is a video with standard resolution. Therefore, the resulted resolution of the dynamic object is lower than that of the static object. Also, based on the requirement of the movable viewpoint, the resolution needs to be compromised with the processing time for real-time display. That is why the resolution of the dynamic object is lower than that of the static object.

The proposed approaches in this paper are to design an anamorphic imaging system for the moving viewer and the major requirement is to fulfil the real-time projection of 3D images to the viewer without delay. In this imaging system, the most complicated processes are to generate the dynamic videos and to locate the viewpoint of the viewer. The process of dynamic videos focuses on the transformation of projecting the 3D image onto the 2D plane and the recognition of the viewer's position is for the viewpoint process. Since prior information can be used to simplify those two computations, the real-time display can be achieved even for the most complex applications according to our experimental results. Therefore, the computation time is not listed in this paper and only explained by words.

5 Conclusions

The traditional perspective anamorphosis can provide a 3-D visual effect without any auxiliary devices, and the techniques of the 3-D display have been widely applied in many fields. Due to the traditional perspective anamorphosis has the limitations of displaying only the static content and requiring a large plane for painting and showing the 3-D effect, this paper presents a new 3-D display approach without attaching any auxiliary devices on the human body and an anamorphic imaging system has been developed for moving viewers by projecting the content onto two orthogonal planes.

The proposed projection-based anamorphic imaging system only employs simple hardware such as the projector, PC, and the camera. Compared to the traditional perspective anamorphosis, our system can provide more advantages: (1) The displayed content can be

the static or the dynamic types, (2) The system allows the viewer to change the viewing position, and (3) The displayed content can be replaced quickly. The experimental results have shown that the proposed system can display static and dynamic 3-D content with high quality and allow the viewer to move. In the future, this system can provide new applications for 3-D display in the advertisement, museums, art galleries, and exhibitions.

Availability of data and material Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest Not applicable.

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